MathEXplainedMAGAZINE

Christmas Math Crossword Solutions

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1. 1	2. 3	3. 0
4. 0	6	3
5. 2	4	8

1 ACROSS

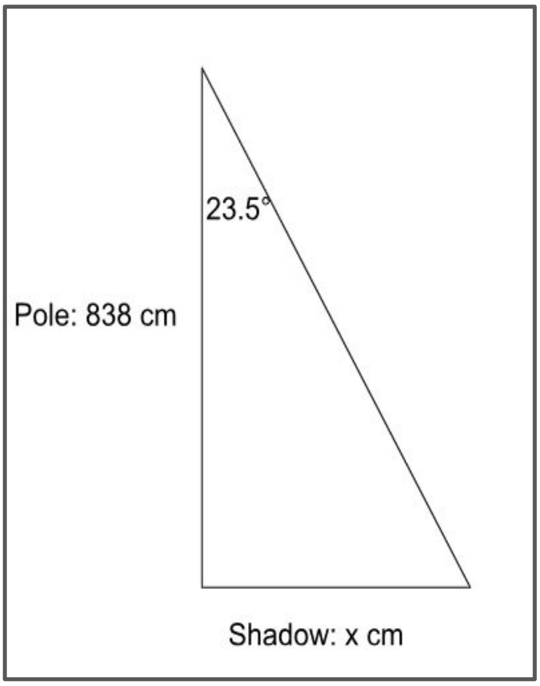
Let x be the number of households that the Grinch ransacks; the total number of presents he steals can then be modeled by the expression (1+x)(250-10x).

$f(x) = -10x^2 + 240x + 250$
$f'(x) = -20x + 240$
Since $f'(12^-) > 0$ & $f'(12^+) < 0$, f(x) reaches a local max at x = 12 – the only extrema on the function. Thus, the number of households that the Grinch should ransack is 250-10*12 = **130**.

1 DOWN

For every set of 100 3 digit numbers X00 - X99 ($X \neq 9$), there are exactly 19 numbers which contain a 9: X09, X19, X29, X39, X49, X59, X69, X79, X89, X90, X91, X92, X93, X94, X95, X96, X97, X98, X99. Thus, there are 19 * 5 = 95 numbers containing 9 from $X \in \{0, 1, 2, 3, 4\}$. When X = 5, there are 7 different numbers containing 9 before 575: 509, 519, 529, 539, 549, 559, 569. Thus, the answer is 95 + 7 = **102**.

2 DOWN

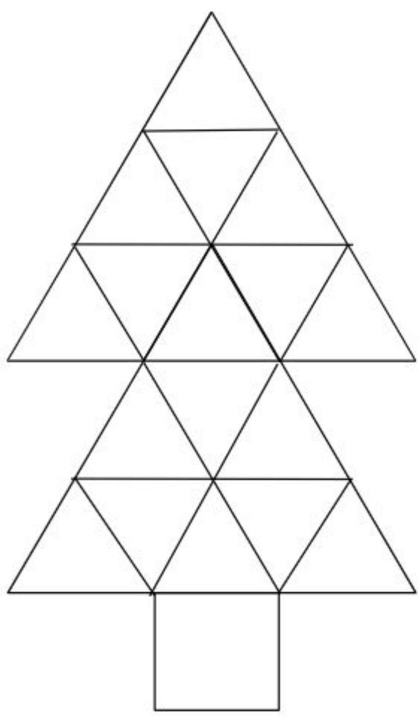


We can begin by constructing a right triangle where the light hits the earth. We can find that the angle at which the light from the sun makes with the pole is 23.5°. From here, we can use trigonometric functions to find the length of the shadow. We have that tan(23.5°) = x/838 cm. Therefore we have x = 838 * tan(23.5°) ≈ **364** cm.

3 DOWN

Denote N = ABCD to be our number, where A,B,C,and D are digits from 0 to 9 and A is not equal to 0. We need to find the smallest possible value of N (and therefore A) such that (AB+CD)^2 = ABCD. Since, AB = 10A+B and CD = 10C+D, we have that (AB+CD)^2 = (10A+10C+B+D)^2 = 1000A+100B+10C+D. Since N is a four digit number, we have that 10A+10C+B+D can be anywhere between 32 and 99 (inclusive). Hence, checking until we find a pair that works, we get that 45 works since 45^2=2025 so 10(2)+10(2)+0+5 = 45 and 45^2 = 2025. Hence, our answer is **2025**.

4 ACROSS



We begin by breaking the tree shape into 17 congruent equilateral triangles. We may count that 5 triangles make up the height of the tree, making the height of each triangle 6/5 cm. From the slides in a 30°- 60°- 90° triangle, we can find the side length for each triangle to be $(6/5)/(\sqrt{(3)}/2)$. The area of each small triangle will be $(½)(4\sqrt{(3)}/5)(6/5) \approx 0.83$ cm^2. The total area of all the small triangles will be 17*0.83 cm^2 ≈ 14.13 cm^2. The area of the trunk of the tree will be $[4\sqrt{(3)}/5]^2 = 1.92$ cm^2. In total the area of the tree is about 16.05 cm^2. The inner circle has a radius of (12/2) - 1 = 5 cm. The total area of the frosting will be the area of the inner circle minus the area of the tree, which is 25π - 16.05 cm^2 ≈ 62.49 cm^2. Taking the ceiling of which is **63** cm^2.

5 ACROSS

Since our polynomial is cubic and monic, let us express it as p(x), which is equivalent to x^3+bx+c+d. Since p(2) = 8, p(p(2) = p(8) = 8. Finally, p(8+8) = p(16) = 8. Hence, p(2)=p(8)=p(16)=8. If q(x) = p(x)-8, the roots of q are 2, 8, 16. Since p is monic, even q is monic, q(x) = (x-2)(x-8)(x-16) = p(x)-8.
p(x) = (x-2)(x-8)(x-16) + 8; p(0) = (-2)(-8)(-16)+8 = -256+8 = -24; p(x)| = **248**